

Ukulele Robot With Mobile Application

Chiu Wing Tung Crystal 1155174499, Cheng Ka Chi 1155175548

Department of Computer Science and Engineering, The Chinese University of Hong Kong

Current version published September 20, 2024.

Index Terms: Ukulele, Robot, Mobile Application.

1. Project Goals

The project aims to create an ukulele robot that can interact with a mobile application. The robot will be able to play the score or chord on remote. The application can also act as accompaniment, to let the user have an real-time interactive experience.

2. Significance of the Project

Integrating robotic into music industry has become increasingly common in recent years. A robotic musical instrument is characterized as a device that autonomously generates music through the utilization of mechanical components, including motors, gears and solenoids [1]. With the incorporation of technology this facilitate the performance, accessibility and even composition of music. The convergence of robotics and musical art has led to the development of diverse types of musical robots, including piano robots, turntable robots, percussion robots, string robots, and wind robots [1].

3. Problem statement

After studying the latest trends in the area of string robots, especially the guitar and ukulele robots. Although there are many recorded resources on the guitar projects, there is less documented information on the ukulele robots making challenging for users to learn about them. Also, most of the current ukulele projects incorporate solenoid mechanisms which attract and release a magnet and in the process create noise which could be avoided in the performance.

4. Proposed Solution

The ukulele robot is a complex system combined with two primary components: hardware and software.

4.1. Hardware

The hardware part include multiple parts to simulate the human interaction with the Ukulele. First, the robot is equipped with multiple arms. The arms are connected to a slider to control the vertical move and a extra motor for horizontal move. The arms are going to precisely press chords and

perform different fingering techniques. This design is to simulate the human playing.

To produce the sound, the robot uses a servo motor integrated with a fan to interact with the strings.

4.2. Software

The software is an additional to the hardware component. The mobile application will allow the user to choose their own songs. It can also let the user to play a specific chord or fingering technique. It can provide an interface to the users on using the self-playing function of the robot.

Integrating with software, the robot can improve the performance of playing music and provide personalized music experience.

5. Proposed Timeline

First Semester		
	Software	Hardware
Week 1	Research and planning	
Week 2	Mobile Design	Design prototype
Week 3		Buy and ship components
Week 4		3D print
Week 5	Fine-tuning for one set of slider	3D print and Adjustment
Week 6 - 8	Communication on multiple slider	Assemble all components
Week 9	Communication on multiple parts	
Week 10 - 14	Mobile app development	
Second Semester		
Week 1 - 2		Auto tuning
Week 3 - 4		Chord playing
Week 5		Fingering playing
Week 6 - 10		AI script detecting
Week 10 - 13		Modification of mobile app

References

- [1] Kapur, A. (2005, September). A history of robotic musical instruments. In ICMC (Vol. 10, No. 1.88, p. 4599).